B.Sc. Hons. (CA&IT) / Data Science Sem-I

[U11E6CMT] COMPUTER MAINTENANCE AND TROUBLESHOOTING

UNIT-2

Basic Troubleshooting Principles and Hardware

Prepared By:

Prof. Pankti Patel

Department of Computer Science Ganpat University

❖Troubleshooting: A Systematic Approach to Problem Solving

➢ What is Troubleshooting?

Troubleshooting is a structured method used to detect, diagnose, and resolve faults in complex systems—especially in computing, electronics, and machinery. It involves gathering relevant information, identifying the cause of the problem, and implementing a suitable solution.

Example: If your laptop won't connect to Wi-Fi, troubleshooting involves checking if Wi-Fi is enabled, verifying the router is working, and checking network settings.

➢ How Does Troubleshooting Work?

Troubleshooting can be applied in various technical environments. Some common areas include:

- **Operating Systems** – e.g., Windows not booting or frequent system crashes
- **Applications** – e.g., a word processor freezing during startup
- **CPUs** – e.g., overheating or failing to start
- **Firewalls** – e.g., blocking legitimate applications from accessing the internet
- **Hard Drives/SSDs** – e.g., data not being saved or drive not recognized
- **Servers** – e.g., not responding to network requests

Example: A firewall may block a file-sharing app. Disabling the firewall temporarily can help verify if it's the cause.

1. Problem Identification
2. Information Gathering
3. Problem Analysis
4. Implementation
5. Solution Testing

1. Problem Identification

- **Observe Symptoms**

Note specific error messages, unusual noises, lights, or performance drops.

Example: The user says the mouse is not moving the cursor. You check the computer screen and notice the pointer is stuck, and the mouse light is off.

- ### Define the Problem

Clearly identify what part of the system is malfunctioning.

Example: The mouse is not responding at all. Since it's a wired mouse, this could be a problem with the **mouse itself**, the **USB port**, or the **computer's settings**.

2. Information Gathering

- ### Check Connections

Ensure all cables, power cords, and internal components are securely connected.

Example: You see the mouse is plugged into a USB port on the front of the CPU. You unplug it and try a different USB port on the back—and the mouse light turns on!

- ### Review Recent Changes

Identify any hardware changes or updates made recently.

Example: The user says they moved the CPU to clean the desk. This could have caused a loose connection or damage to the front USB port.

- ### Consult Documentation

Check the system or motherboard manual for diagnostic codes, beep sequences, or power diagrams.

Example: You check the PC manual and it shows how to test USB ports. You confirm that the front ports may not work unless enabled in BIOS (which was not changed), so the problem is likely physical.

3. Problem Analysis

Ask diagnostic questions to narrow the problem:

- What are the symptoms?
- When did the problem start?
- Where does it happen?
- Under what conditions?
- Can it be repeated?

Example Analysis:

- Symptoms: Mouse is not working
- When: After moving the CPU
- Where: On the front USB port
- Conditions: Only on the front ports
- Reproducible?: Yes — the mouse works only when plugged into the back

This suggests the front USB ports are faulty or loose.

4. Implementation

- **Execute the Fix**: Implement the selected solution, such as replacing a faulty component, updating firmware, or reconfiguring the hardware setup.
- **Ensure Compatibility:** Verify that any replacement parts or updates are compatible with the existing system.
- **Follow Safety Procedures**: Always follow proper safety protocols, like grounding yourself before handling sensitive components, to prevent damage or injury.

5. Solution Testing

- **Test the Hardware**: After implementing the solution, test the hardware to ensure the problem is resolved. This might involve running the system through various scenarios or stress tests.
- **Monitor for Stability**: Observe the system over time to ensure the problem does not reoccur and that the fix did not introduce new issues.
- **Seek User Feedback:** If the hardware is used by others, get their feedback to confirm that everything is working as expected.

❖ Common Computer Problems

1) The Computer Won't Start

Problem:
The computer shuts off suddenly or doesn't turn on at all.

Possible Cause:
- Faulty power supply
- Loose power cable
- Dead battery (in laptops)

Solution:
- Make sure the computer is plugged into the power socket properly.
- Test the power outlet with another device to check if it's working.

- If the outlet is fine, try using a different power cable or adapter.

2) The Screen is Blank

Problem:
The computer turns on, but nothing appears on the screen.

Possible Cause:
- Loose cable between monitor and CPU
- Monitor not powered on
- Graphics card issue

Solution:
- Ensure the monitor's power cable is plugged in and the monitor is turned on.
- Check the connection between the monitor and the CPU.
- Try a different monitor or cable to test.

3) Operating System or Software Not Working Properly

Problem:
The system or an application is slow, freezing, or not responding.

Possible Cause:
- Temporary software glitch
- Virus or malware
- Corrupt files

Solution:
- Restart the computer.
- Run a virus scan using a trusted antivirus.
- Keep your software updated.

4) Windows Won't Boot

Problem:
The system fails to load Windows or shows a boot error.

Possible Cause:
- Corrupt system files
- Hardware failure

- Improper shutdown

Solution:

- Use a Windows recovery disk or USB to repair or reinstall Windows.
- Check the BIOS to make sure the hard drive is detected.

5) The Screen is Frozen

Problem:
The computer becomes unresponsive; keyboard and mouse don't work.

Possible Cause:
- Too many programs open
- Software crash
- Driver issues

Solution:
- Press and hold the power button until the system shuts down.
- Restart the PC and close unnecessary programs.
- Run disk cleanup and update drivers.

6) Computer is Slow

Problem:
System performance is unusually slow.

Possible Cause:
- Too many startup programs
- Low disk space
- Malware or spyware [Malware = **malicious software** designed to harm your system or steal data. Spyware = a type of malware that secretly monitors your activity and sends information to attackers.]

Solution:
- Delete unwanted files and programs.
- Use Disk Cleanup and Defragment tools.
- Install antivirus, anti-malware, and perform regular scans.

7) Strange Noises

Problem:
You hear clicking, grinding, or buzzing noises.

Possible Cause:
- Failing hard drive
- Loose or damaged fan
- Faulty internal components

Solution:
- Back up your data immediately if you hear clicking (possible hard drive failure).
- Open the case and check for dust buildup or loose fans.
- Replace noisy fans.

8) Slow Internet

Problem:
Web pages take too long to load or downloads are very slow.

Possible Cause:
- Too many temporary files
- Browser extensions or cookies
- Poor Wi-Fi signal

Solution:
- Clear browser cache, cookies, and temporary files.
- In Windows, type %temp% in the search bar and delete unnecessary files.
- Restart your modem/router and reduce the number of connected devices.

❖ Introduction to Diagnostic Tools

- Diagnostic tools in computers are specialized software and hardware used to identify and troubleshoot issues within a computer system.
- They analyze hardware and software components to pinpoint problems and improve system performance.

➢ Types of Diagnostic Tools:

Hardware Diagnostics	Check physical parts (CPU, RAM, Hard Drive, etc.). Example: BIOS tools, diagnostic cards.
Software Diagnostics	Check system software, apps, and drivers.

	Example: Event Viewer, Resource Monitor.
Performance Monitoring Tools	Track CPU, memory, disk, and network usage.
Troubleshooting Tools	Fix specific issues (e.g., network problems). Example: Network Troubleshooter.
Memory Diagnostics	Test RAM for errors. Example: Windows Memory Diagnostic.
Disk Diagnostics	Scan and repair hard drive errors. Example: CHKDSK.

- **Examples of Common Diagnostic Tools:**

Task Manager/Resource Monitor: Provides information about running processes, resource usage (CPU, memory, disk), and network activity.

Event Viewer: Logs system events, errors, and warnings, providing insights into potential problems.

Performance Monitor: Allows you to track various system performance counters in real-time or log them for later analysis.

Windows Memory Diagnostic: Tests the computer's RAM for errors.

CHKDSK (Check Disk): Scans hard drives for errors and attempts to repair them.

Device Manager: Provides information about installed hardware devices and allows you to troubleshoot driver issues.

CPU-Z: Provides detailed information about the computer's CPU, including its core specifications, clock speeds, and cache.

Speccy: Offers detailed information about the computer's hardware components, including the CPU, motherboard, RAM, graphics card, and storage devices.

HD Tune: Tests the health and performance of hard drives and SSDs.

Open Hardware Monitor: Monitors temperatures, voltages, and fan speeds of various hardware components.

WiFi Analyzer: Helps diagnose and optimize WiFi network connectivity.

➢ BIOS/UEFI

The **BIOS (Basic Input/Output System)** or its modern successor, **UEFI** (**Unified Extensible Firmware Interface**), is firmware that runs when you first turn on your computer. It is stored on a chip on your motherboard and is the first piece of software to load.

Difference :

- **BIOS** = Old style (used in older PCs).
- **UEFI** = New style (used in modern PCs, faster and safer).

- **How to access it:** You typically press a specific key (like `F2`, `F10`, `Del` or `Esc`) immediately after powering on your computer.

➔ What BIOS/UEFI Does:

1. **Checks hardware** → Makes sure RAM, CPU, keyboard, screen, etc. are working (this is called **POST – Power On Self Test**).
2. **Find the storage drive** → Hard disk/SSD/USB.
3. **Loads the operating system** → Starts Windows, Linux, etc.
4. **Lets you change settings** → You can enter BIOS/UEFI setup (by pressing keys like F2, DEL, ESC at startup) to change things like boot order, time/date, or enable/disable devices.
5. **Controls how the computer boots** → Decides if the computer should boot from Hard disk, USB, or DVD.
6. **Provides basic drivers** → Gives simple instructions so that keyboard, screen, and storage work before the full operating system loads.
7. **Security features (UEFI especially)** → Password protection, Secure Boot (prevents viruses/malware from loading before Windows starts).
8. **Supports updates** → BIOS/UEFI itself can be updated to fix bugs or support new hardware.

➢ System Information Utilities

These are software tools that run within your operating system and provide detailed information about your hardware and software. They are useful for quickly gathering a complete overview of your system's configuration without restarting the computer.

How to access them:

- **Windows:** Use the `msinfo32` command (System Information)

- **macOS:** Go to the Apple menu > About This Mac > System Report.

➔ What they show you:

1. **Basic computer details** → Computer name, model, manufacturer.
2. **Hardware details** → Processor (CPU), Memory (RAM), Storage (Hard disk/SSD), Graphics card.
3. **Software details** → Windows version, installed programs, drivers.
4. **Network details** → Internet adapter, IP address.
5. **Performance & health** → How much RAM is used, how much storage is free.

❖ Troubleshooting Common PC Hardware Problems

1. Blank Monitor

- Check if the monitor is properly plugged in.
- Make sure the power supply is working.
- Push and reconnect the video cable properly.
- Try restarting the PC after checking connections.

2. Mouse Problems

- Make sure the mouse is connected properly (USB or wireless).
- Check if batteries are charged (for wireless mouse).
- Clean the mouse sensor or ball.
- Try using the mouse on a different surface.

3. Jumpy Mouse

- For ball mouse: Open it, remove the ball, clean it.
- For optical mouse: Clean the sensor at the bottom.
- Try using a mouse pad for better performance.

4. USB Camera Not Recognized

- Turn on the camera before connecting it.
- Try connecting to a different USB port.
- Check if the camera driver is installed.

- Restart the PC and try again.

5. Smartphone Won't Sync with PC

- Close all running apps on both devices.
- Ensure your phone's sync software is installed.
- Use the original USB cable.
- Try syncing after restarting both devices.

6. Keyboard Problems

- Check the keyboard connection (wired/wireless).
- Try using another keyboard to test the issue.
- Clean stuck or unresponsive keys.
- If letters appear jumbled, check language settings.

7. Power Cord Problems

- Make sure the power cord is properly plugged in.
- Check for cuts or damage on the cord.
- Try a different power outlet.
- For laptops, ensure the battery is charging.

8. Motherboard Problems

- Problems could be due to RAM, BIOS, or CPU.
- Check for error beeps or blinking lights.
- Remove and reinsert RAM if needed.
- If nothing works, consider replacing the motherboard.

9. Strange Noises

- Could be from the hard drive or fan.
- Back up your data immediately.
- Clean or replace noisy fans.
- Seek help if the hard drive is clicking.

10. Overheating

- Turn off the PC and let it cool.
- Check if the fan is working properly.
- Clean dust from vents and fans.

- Use the computer in a cool, open space.

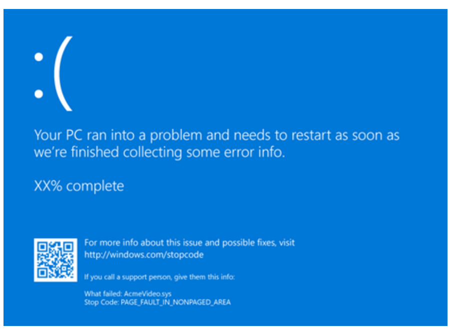
11. Slow Internet

- Clear temporary files: Press "Windows + R, type %temp%," hit Enter, and delete files.
- Restart your router.
- Scan for viruses or malware.
- Avoid too many open browser tabs.

12. Insufficient Memory

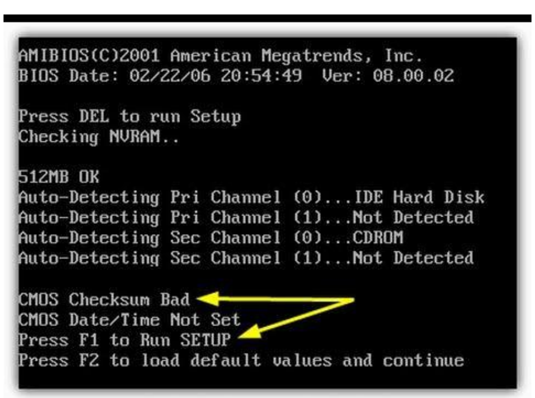
- Too many apps open? Close unused ones.
- Add more RAM to your system if needed.
- Delete or move unnecessary files.
- Use lighter software alternatives.

13. Blue Screen of Death



- Caused by hardware or driver problems.
- Restart your computer.
- Update drivers or uninstall recent changes.
- Seek help if it happens often.

14. CMOS Error(Complementary Metal-Oxide Semiconductor)



- CMOS stores time/date and hardware settings.
- Replace the CMOS battery on the motherboard.
- Make sure to insert the same type of battery.
- Reset BIOS settings to default after replacing.

❖ Safe handling practices for computer components

 Handling computer parts safely is very important because they are delicate and can be damaged by **static electricity, rough handling, or dust**.

1. Prevent Static Electricity (ESD)

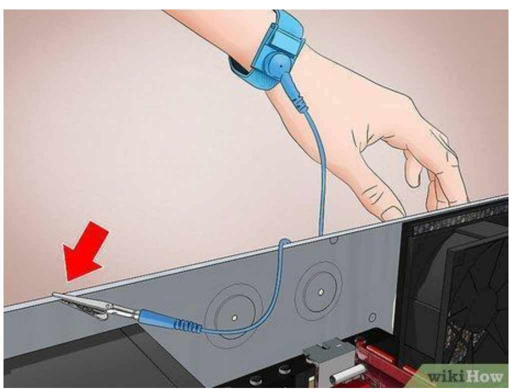
 This is the most crucial part! Our bodies can build up static electricity, especially on dry days or if we've been shuffling our feet on carpet. When you touch a sensitive computer part, that static can jump and fry the component.

➢ **Ground yourself**

- Touch a metal part of your computer case before handling parts.
- This removes (discharges) any extra electricity from your body.

➢ **Use an ESD wrist strap**

- Looks like a small band you wear on your wrist.
- One end connects to your computer case or a special mat.
- It makes sure your body's electricity goes away safely.



 Work on a clean, clear surface: Avoid working on carpet, which can generate a lot of static. A wooden table or an anti-static mat is ideal.

2. ✋ Handle with Care

 Hold by the edges: When picking up circuit boards (like a motherboard, RAM, or graphics card), always hold them by their edges. Never touch the gold connectors, chips, or tiny parts on the board. Oils from your fingers can cause problems.

- **Don't force anything:** Connectors are designed to fit in a specific way. If something doesn't slide in easily, double-check that it's oriented correctly. Forcing it can bend pins or break the component.
- **Be gentle with cables:** When disconnecting cables, pull on the connector itself, not the wire.

3. 🔌 Power Safety

- **Unplug everything:** Always turn off your computer and unplug it from the wall before opening the case or touching any internal components. This prevents electric shock.
- **Never open the power supply unit (PSU):** The power supply can hold dangerous electrical charges even when unplugged. If it's faulty, replace it; don't try to repair it.

4. 🧹 Keep it Clean

- **Work in a clean area:** A clean workspace prevents dust and debris from getting into sensitive parts.
- **No liquids:** Never use liquid cleaners directly on computer components. A slightly damp cloth (with water, not harsh chemicals) can be used on the outer case, but keep liquids far away from internal parts.